Density and Goldstone emission of an $\mathbb{RP}^3 \simeq \mathrm{SO}(3)$ global-string network relative to U(1), and a massive-bath systematic of the emission estimator

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Draft, October 17, 2026

Abstract

We simulate networks of global $\mathbb{Z}_2$ strings whose vacuum manifold is $\mathbb{RP}^3 \simeq \mathrm{SO}(3)$, produced by a real symmetric-tensor order parameter, and compare them with U(1) axion-string networks evolved in the same code, with the same Goldstone decay constant and core mass, at 576^3 (U(1) control also at 768^3, where it passes preregistered scaling gates). The string-density ratio is $R_\xi \simeq 1.6$ for every protocol tested: sparse and dense starts, with and without draining the initial massive radiation, and networks started late and dense that approach the attractor from above, which bracket it to $R_\xi \in [1.57, 1.68]$ at $\ell = \ln(m_r/H) = 5.61$. We identify a systematic of the standard kinetic Goldstone-emission estimator: the conversion of massive radiation left by the initial state into Goldstones is counted as string emission. It is 19% ($\mathbb{RP}^3$) and 28% (U(1)) of the measured emission for sparse starts and more than half for dense ones; for U(1) two independent methods, a drain of the massive modes and a direct measurement of the radial-to-Goldstone transfer, agree. With the bath drained, the emission ratio is $R_\Gamma = 1.56 \pm 0.09\,(\mathrm{stat}) \pm 0.08\,(\mathrm{sampling})$, independent of the initial density within $\approx 7\%$, and the emission per unit string length is within 12% of U(1) from both sides of the attractor. The spectral slope of the $\mathbb{RP}^3$ emission at $\ell = 5$ is $q = 0.87 \pm 0.06$ with our estimator, but an independent re-implementation shows that the convention used to map the physical fitting window onto lattice momenta shifts it by up to 0.07; we therefore quote q as compatible with 1. All verdict-bearing analyses were preregistered and are reproducible from the released data.

1 Introduction

Networks of global strings radiate Goldstone bosons, and in the post-inflationary axion scenario that radiation sets a large part of the dark-matter abundance and its phase-space distribution [1–7]. Almost all of that work concerns U(1) strings. Non-abelian vacuum manifolds give strings with discrete charges, which can annihilate in pairs rather than unwind, and have several massive modes in the core. Scaling of $\mathrm{SU}(N)/\mathbb{Z}_N$ global-string networks was established recently [8]; their emission was not measured.

Here we measure, in the same code and with matched decay constant and core mass, the string density and the Goldstone emission of a network with vacuum manifold $\mathbb{RP}^3 \simeq \mathrm{SO}(3) \simeq \mathrm{SU}(2)/\mathbb{Z}_2$ relative to a U(1) network. The model arises in a toy cosmology in which the emission feeds a pseudo-Goldstone dark-matter component [9], but the results below do not depend on that model. Three results are, to our knowledge, new: (i) the density ratio R_ξ and its robustness, including an approach to the attractor from above; (ii) a quantified systematic of the kinetic emission estimator, the conversion of the initial massive bath, which is larger for vacua with more massive modes; (iii) the bath-free emission ratio R_Γ and spectral slope q, with an estimator systematic for q found by an independent audit.

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Every analysis that yields a verdict was preregistered, with thresholds and verdict logic fixed before the data, and the measurements behind the main results were audited by an independent re-implementation (§5).

2 Model and lattice set-up

The order parameter is a real symmetric 4×4 matrix Q with potential

$$U(Q) = \tfrac{\lambda}{4}(\mathrm{Tr}\, Q - 1)^2 + \tfrac{\lambda_P}{4}\mathrm{Tr}\big[(Q^2 - Q)^2\big], \tag{1}$$

invariant under $Q \to OQO^T$, $O \in \mathrm{O}(4)$. Its vacua are rank-one projectors $Q = nn^T$, $n \in S^3$, with n and $-n$ identified, so the vacuum manifold is $\mathbb{RP}^3$: $\pi_1 = \mathbb{Z}_2$ gives strings, $\pi_3 = \mathbb{Z}$ gives textures, and there are no domain walls. Around the vacuum there are three Goldstone modes and seven massive modes (a degenerate quintet and singlet of mass m_r, and a heavier trace mode). The U(1) control is a complex scalar with $V = \frac{\lambda_u}{4}(|\phi|^2 - v^2)^2$, $v^2 = f^2 = \frac{1}{2}$ and $m_r^2 = 0.24$, equal to the quintet mass of $\mathbb{RP}^3$; with these choices the string tension normalisation $\mu = \pi f^2 \ell$ is the same.

Both models are evolved in a radiation-dominated background with physical cores (no Press–Ryden–Spergel fattening), with a symplectic leapfrog in conformal time, a 7-point Laplacian, $dx = 1$, $d\tau = 0.25$ and $a = s\tau$, $s = 1/(m_r \tau_{\mathrm{end}})$, so that $m_r a\, dx \le 1$ and $\ell = \ln(m_r/H) = \ln(\tau^2/\tau_{\mathrm{end}})$ reaches $\ln(N/2)$ (5.66 at 576^3, 5.95 at 768^3). Initial conditions are Gaussian random fields with a correlation length chosen for a target initial density ξ_i; for $\mathbb{RP}^3$, n is the principal eigenvector of a random symmetric matrix and $Q = nn^T$.

Observables. Strings are detected on plaquettes from the $\mathbb{Z}_2$ holonomy $\prod \mathrm{sign}(n_i \cdot n_j) = -1$ ($\mathbb{RP}^3$) or the winding (U(1)), and $\xi = L\tau^2/(4V)$ in comoving units. Free Goldstone radiation is measured with the estimator of Ref. [1]: twice the kinetic energy tangent to the vacuum manifold, $\rho_a = \langle 2K_G \rangle$, averaged over cells farther than $2.5/m_r$ from any pierced plaquette. The emission rate is $\Gamma = Ha^{-4} d(a^4 \rho_a)/d\ln\tau$, by centred differences between snapshots, normalised as $\varepsilon = \Gamma/(8\pi H^3 f^2 \ell \cdot 0.24\,\ell)$; $R_\Gamma = \varepsilon_{\mathbb{RP}^3}/\varepsilon_{\mathrm{U}(1)}$ and $R_\xi = \xi_{\mathbb{RP}^3}/\xi_{\mathrm{U}(1)}$ are ratios of seed means over the window $\ell \in [\ell_{\mathrm{end}} - 1.3, \ell_{\mathrm{end}} - 0.05]$, with errors from a bootstrap over seeds. The spectral slope q is fitted to the instantaneous emission $F(k) = a_2^4\, dE_2(k) - a_1^4\, dE_1(k)$ between consecutive snapshots, $F \propto k^{-q}$ for $10H < k/a < m_r/4$ with at least eight bins, and $q(\ell) = q_0 + q_1(\ell - 5)$ is fitted for $\ell \ge 4.6$.

Validations. The linear spectrum around the vacuum is reproduced to 6×10^{-5}; the detector is exact on straight strings; with no strings the comoving Goldstone energy is conserved to 2.9% and no string is counted; the estimator gives the same value for the same Goldstone packet in both models. A dedicated resolution test with matched initial conditions (the same physical field on 192^3, 256^3 and 320^3 grids at fixed LH, so that $m_r a\, dx = 1$, 0.75 and 0.6 at the end of the run, three seeds, $\ell \le 4.56$) changes R_ξ by $-0.26 \pm 0.05\%$ between $m_r a\, dx = 1$ and 0.6, and by -0.2% at $m_r a\, dx = 0.5$ (384^3). For R_Γ the same test first suggested a small resolution dependence of the local estimator ($+1.6\%$ at 0.6, $+4\%$ at 0.5), but a repetition of one seed with densely sampled snapshots showed it to be a sampling effect: with dense sampling the local and integral estimators agree to $< 1\%$ and change by -2% and -0.8% between $m_r a\, dx = 1$ and 0.6, while with the ≈ 30 snapshots per run used for all production networks the local estimator departs from its densely sampled value by $+1.4\%$ and $+4.9\%$, and the integral one by up to -12% (the 0.5 case was not repeated). We therefore use the local estimator only and add a $\pm 5\%$ sampling systematic to R_Γ. A box with 4.6 times more Hubble volumes changes R_ξ and R_Γ by $-0.6 \pm 5.5\%$ and $+0.7 \pm 4.9\%$.

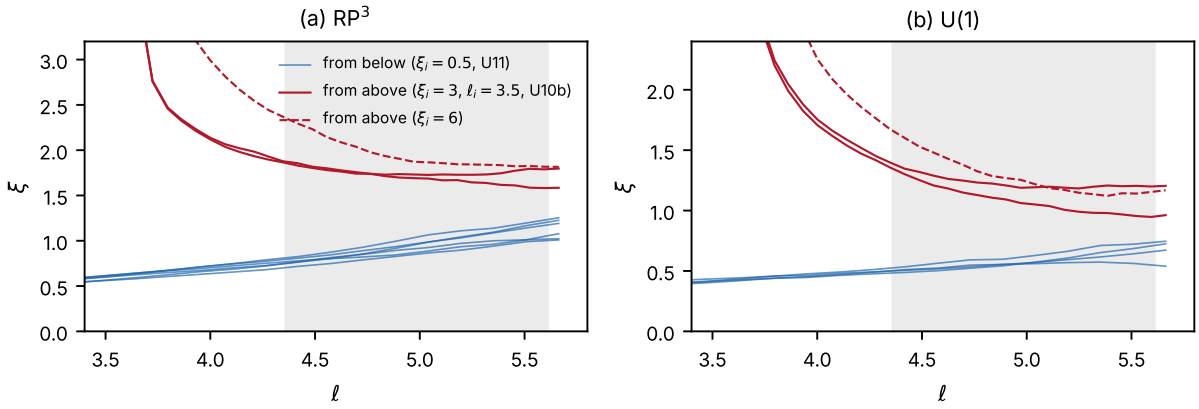


Figure 1: The scaling attractor bracketed at 576^3. Blue: drained networks starting sparse ($\xi_i = 0.5$), approaching from below. Red: drained networks started late ($\ell_i = 3.5$) and dense ($\xi_i = 3$ solid, 6 dashed), approaching from above. The shaded band is the analysis window.

3 Results

3.1 U(1) control at the attractor

At 256^3 the U(1) control kept a 31% memory of its initial density; at 768^3 four networks pass the preregistered gates, $\xi_{\mathrm{U}(1)}(5.65) = 0.62 \in [0.5, 2.5]$ and a memory $|\xi(\xi_i{=}2) - \xi(\xi_i{=}0.25)|/\bar{\xi} = 7.6\% < 20\%$. The memory falls with box size (31%, 13%, 7.6% at 256^3, 384^3, 768^3). The $\mathbb{RP}^3$ network, with ten field components, is measured at 576^3 against U(1) at the same size; the 576^3 block inherits the attractor gates of the 768^3 control.

3.2 Density ratio

With six $\mathbb{RP}^3$ and four U(1) seeds starting at $\xi_i = 0.5$, $R_\xi = 1.617 \pm 0.053$. The $\mathbb{RP}^3$ density grows about three times faster than the U(1) one in the window ($d\xi/d\ell = 0.31$ against 0.10). Dense starts ($\xi_i = 2$) give the same density and slope within 5% ($R_\xi = 1.58 \pm 0.09$), but all collapse below their final density at $\ell \approx 3.3$–4.3 and then grow, so they do not test an approach from above. We therefore started networks late and dense, at $\ell_i = 3.5$ with $\xi_i = 3$ and 6, after the generic collapse (Fig. 1). They decrease through the whole window in both models ($d\xi/d\ell = -0.20$ for $\mathbb{RP}^3$, -0.26 for U(1)), while the sparse starts increase ($+0.29$, $+0.14$). At $\ell = 5.61$ the scaling density is bracketed to $[1.12, 1.73]$ for $\mathbb{RP}^3$ and $[0.67, 1.10]$ for U(1), and the ratio is the same from both sides, $R_\xi \in [1.57, 1.68]$. The growth of sparse networks is therefore consistent with an approach from below, as argued for U(1) in Refs. [7, 10]; our window cannot distinguish a constant attractor from a slowly growing one.

3.3 The massive-bath systematic

Dense starts emit more than sparse ones ($\times 1.85$ for $\mathbb{RP}^3$, $\times 1.52$ for U(1)), with a bluer spectrum, and keep 20–29% more massive radiation outside the cores at the end of the window. Since the density is the same, the excess cannot come from the strings. The simplest reading is that the initial state leaves a bath of massive radiation that keeps converting into Goldstones inside the window, and that the kinetic estimator counts this conversion as emission.

We tested this in two independent ways. *Drain:* until $\ell_d = 3.8$ the massive part of the velocity (normal to the vacuum manifold) is damped at the physical rate m_r, only where the order parameter is close to the vacuum, so cores are untouched; afterwards the evolution is the physical one. *Exchange:* for U(1) we measure the local transfer from the radial mode to the Goldstone sector, $X = 2(\dot{r}/r)(e_{G,\mathrm{grad}} - e_{G,\mathrm{kin}})$, which follows from $\partial_t e_G = X + \nabla \cdot (r^2 \dot{\theta} \nabla \theta)$ and

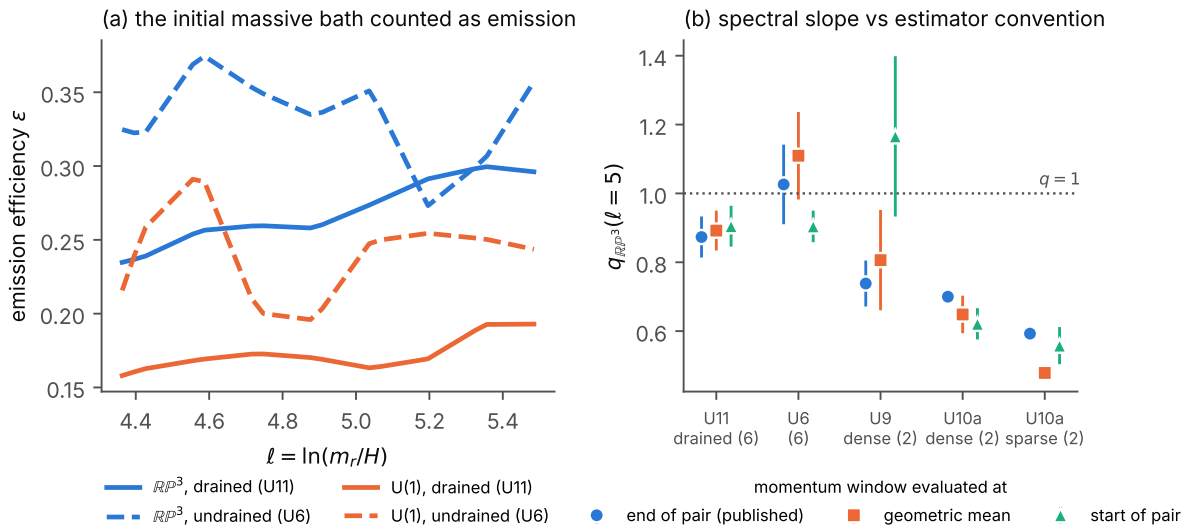


Figure 2: (a) Emission efficiency (seed means) of the sparse networks with the initial massive bath drained (solid) and of their undrained twins (dashed). (b) Spectral slope $q_{\mathbb{RP}^3}(\ell = 5)$ with bootstrap errors for each data set, with the physical fitting window $[10H, m_r/4]$ converted to lattice momenta using a and H at the end, the geometric middle or the start of each snapshot pair.

closes the Goldstone energy budget of a string-free box to 12% (its plane-by-plane analogue for $\mathbb{RP}^3$ fails that validation and is used only qualitatively).

For the dense U(1) network the drain leaves 43% of the emission, and the undrained exchange attributes 45% to the strings: two methods agree that more than half of the dense-start "emission" is bath conversion. We then repeated the six $\mathbb{RP}^3$ and four U(1) sparse networks with the same seeds and the bath drained, so that each drained network has an undrained twin (Fig. 2a). Seed by seed the drain leaves 0.81 ± 0.05 of the $\mathbb{RP}^3$ emission and 0.72 ± 0.04 of the U(1) one: the bath was 19% and 28% of the sparse-start emission. Because it inflated U(1) more, the undrained $R_\Gamma = 1.39 \pm 0.06$ was biased low. In the drained U(1) networks the measured exchange is below 2.5% of the emission. Within the window the bath contribution fades: the drained-to-undrained emission ratio of $\mathbb{RP}^3$ rises by 0.28 ± 0.08 per unit ℓ (U(1): 0.04 ± 0.03), while the massive radiation of the network itself, which the drain does not touch, grows.

The drain removes a lattice artefact rather than part of the cosmological production: the massive modes decay at tree level well before the late epochs relevant for cosmology, into hard Goldstones that redshift as radiation. The systematic is, however, intrinsic to the kinetic estimator at accessible ℓ, and it is larger for vacua with more massive modes.

3.4 Bath-free emission and spectrum

With the bath drained (six $\mathbb{RP}^3$ and four U(1) seeds, 576^3),

$$R_\Gamma = 1.56 \pm 0.09\,(\text{stat}) \pm 0.08\,(\text{sampling}), \qquad R_\xi = 1.61 \pm 0.06, \tag{2}$$

that is $R_\Gamma = 1.56 \pm 0.12$ in total, and $R_\Gamma = 1.54 \pm 0.09\,(\text{stat})$ when the measured U(1) exchange is subtracted. At 512^3 with two seeds per protocol the drained ratio is 1.63 ± 0.02 for sparse and 1.76 ± 0.09 for dense starts, a protocol gap of 7% against 20% without draining. Per unit string length the $\mathbb{RP}^3$ network emits within 12% of U(1) from both sides of the attractor ($R_\Gamma/R_\xi = 0.97$ from below, 0.87 from above): there is no "silent" non-abelian network.

For the spectral slope the drained networks give $q_{\mathbb{RP}^3}(\ell = 5) = 0.87 \pm 0.06$, decreasing with ℓ ($dq/d\ell = -0.32 \pm 0.18$), with the window converted to lattice momenta at the end of each snapshot pair. An independent re-implementation reproduces this number exactly, but finds that

the conversion convention matters (Fig. 2b): at the geometric middle or the start of the pair the same data give 0.89 ± 0.06 and 0.91 ± 0.06, and a fit weighted by F gives 0.94 ± 0.04. The spread is comparable to the statistical error, so the preregistered classification $q < 1$ at 2σ is not robust, and we quote q as compatible with 1. With two seeds the convention moves q by far more than its bootstrap error (by up to 0.4 for the dense starts), so earlier two-seed values should be read as $q \approx 0.5$–1.2. The sensitivity indicates that the emission spectrum is not a clean power law over the small range $10H < k/a < m_r/4$ available at $\ell \approx 5$.

4 Discussion

Comparison with other work. Yamada [8] finds scaling $\mathrm{SU}(2)/\mathbb{Z}_2$ networks with $\zeta \simeq 1.5$ (fixed comoving core) and 2.0 (physical core, slowly growing), and a density roughly proportional to N^2-1. Our same-code ratio to U(1), $R_\xi \simeq 1.6$, is consistent with a denser non-abelian network; the U(1) control density, $\xi \simeq 0.6$ at $\ell \simeq 5.6$, is on the low side of published values at this ℓ [2–4], as expected from our sparse initial conditions, and the late dense starts bracket it. Our dynamic range ($\ell \leq 6$) is below the state of the art for U(1) ($\ell \approx 8$–9) [7, 10], so our U(1) runs are a control, not a contribution to the scaling dispute.

Implications of the bath systematic for U(1). The kinetic estimator is standard in U(1) studies. A 28% contamination at $\ell \lesssim 6$ for sparse starts, and more than half for dense ones, suggests that emission rates and spectra measured at small ℓ should be checked against a drained or exchange-corrected estimator, especially where the initial state is far from the vacuum. The effect fades with ℓ, so large-ℓ simulations should be less affected.

Limitations. (i) The $\mathbb{RP}^3$ runs reach $\ell = 5.66$; nothing here extrapolates to cosmological $\ell \approx 70$. (ii) The drain is applied until a fixed $\ell_d = 3.8$ and its own bias on the network is bounded only indirectly (R_ξ changes by $< 1\%$). (iii) The exchange X is quantitative only for U(1). (iv) The slope q depends on the estimator convention, as shown above.

5 Preregistration, audit and reproducibility

Each analysis was written as a dated specification (thresholds, windows, estimators, verdict logic) before the corresponding networks were produced, and later changes are dated amendments. The preregistered verdicts are: attractor gates passed for the U(1) control at 768^3; genuine growth within the window for sparse and dense starts; an indeterminate verdict for the first drain test at 512^3 (a residual massive excess); a clean emission ratio in the allowed zone and $q < 1$ for the drained 576^3 networks; and a bracketed attractor in both models for the late dense starts. An independent audit re-implemented the estimators from the specifications alone, without reading the analysis code, and reproduced every density, emission and drain number of the production networks to $\leq 10^{-4}$ relative (the later resolution and sampling tests were not audited); it found the q convention dependence reported above and a definitional difference in the per-seed exchange fraction with no effect on R_Γ. Code, network summaries, specifications, audits and figure scripts are public [9].

6 Conclusions

A global-string network with vacuum manifold $\mathbb{RP}^3 \simeq \mathrm{SO}(3)$ is ≈ 1.6 times denser than a U(1) network with the same decay constant and core mass, robustly across protocols and from both sides of the attractor, and per unit length it radiates Goldstones within $\approx 10\%$ as efficiently, giving $R_\Gamma = 1.56 \pm 0.12$ (statistical and snapshot-sampling errors combined) once the initial massive bath is removed. The bath, counted as emission by the standard kinetic estimator, is a

sizeable systematic at accessible ℓ (19–28% for sparse starts, more than half for dense ones). The emission spectrum is close to scale invariant, with a slope at $\ell = 5$ that is compatible with 1 once the estimator convention is accounted for.

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